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22MA101- ENGINEERING MATHEMATICS- I
UNIT I – MATHEMATICSL MODELING OF LINEAR FUNCTIONS- METHOD OF LEAST SQUARES

1.1 General forms of straight Lines:

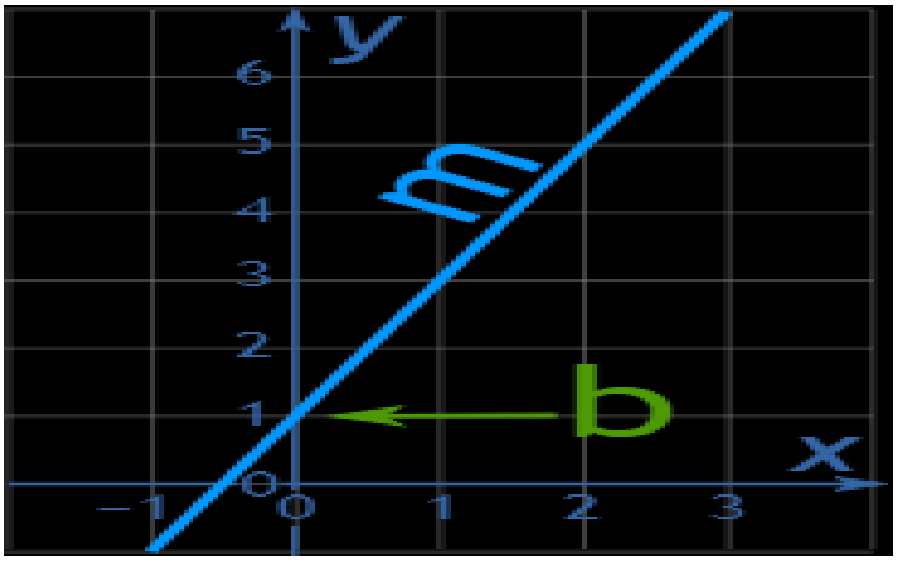
The following table gives the Forms of Linear Equations.

Forms for the Equation of a Line		
Slope-Intercept	$y = mx + b$	m is the slope b is the y-intercept
Point-Slope	$y - y_1 = m(x - x_1)$	m is the slope (x_1, y_1) is a point on the line
Standard Form	$ax + by = c$	a is positive
Intercept Form	$\frac{x}{a} + \frac{y}{b} = 1$	a is the x-intercept b is the y-intercept
Vertical	$x = a$	Vertical line with a as the x-intercept
Horizontal	$y = b$	Horizontal line with b as the y-intercept

1.2 Slope–Intercept Form

y = mx + b
where m is the slope or Gradient of the line and b is the y-intercept.

The y-intercept is the y-coordinate of the location where line crosses the y axis. This is the point when x = 0 and y = b. Vertical lines, having undefined slope, cannot be represented by this form. The slope-intercept form is useful when we are given the slope and y-intercept of a line and we need to write an equation for the line. It is also useful because we can read the slope and y-intercept from the equation. Often, when we are given equations in other forms, we can rewrite it in slope-intercept form to get the slope and y-intercept. The slope-intercept form is also useful when we need to draw the line on a graph.



Example: $y = 2x + 1$

- Slope: **m = 2**
- Intercept: **b = 1**

1.3. Linear Equation in Point Slope Form:

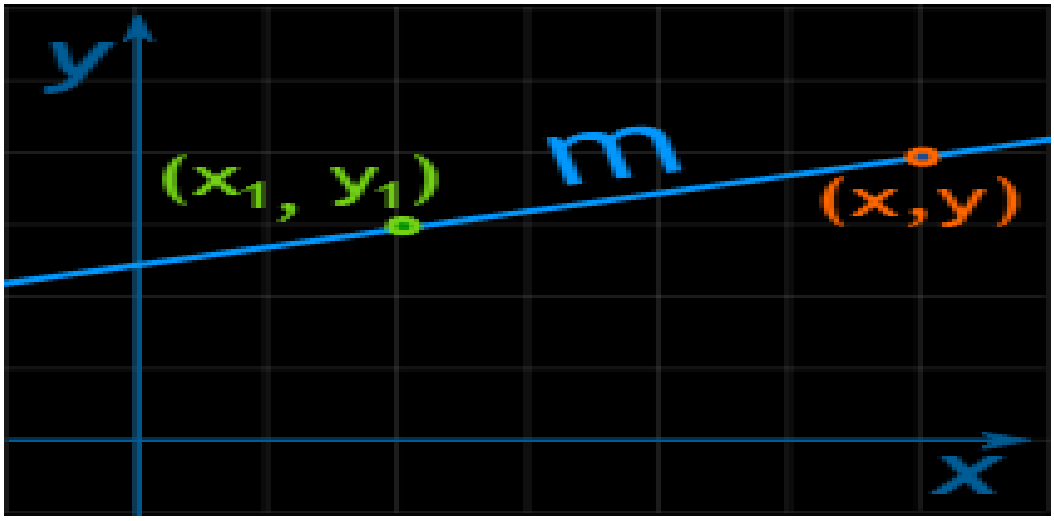
$$y - y_1 = m(x - x_1)$$

where m is the slope of the line and (x_1, y_1) is any point on the line. The point-slope form shows that the difference in the y-coordinate between two points on a line is proportional to the difference in the x coordinate. The proportionality constant is the slope of the line, m. The point-slope form is useful when we are given a point on a line and the slope and we need to get the equation of the line.

Example: $y - 3 = (¼)(x - 2)$

It is in the form $y - y_1 = m(x - x_1)$ where:

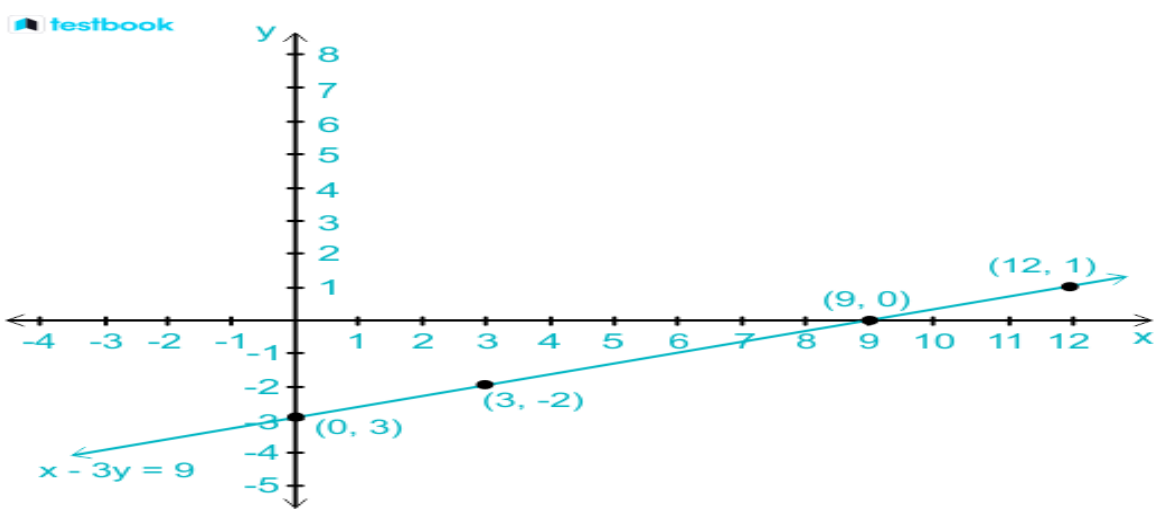
- $y_1 = 3$
- $m = ¼$
- $x_1 = 2$



1.4 Intercepting form of lines:

A line that is not parallel to an axis and does not pass through the origin cuts the axes in two different points. The intercept values x_0 and y_0 of these two points are nonzero, and an equation of the line is

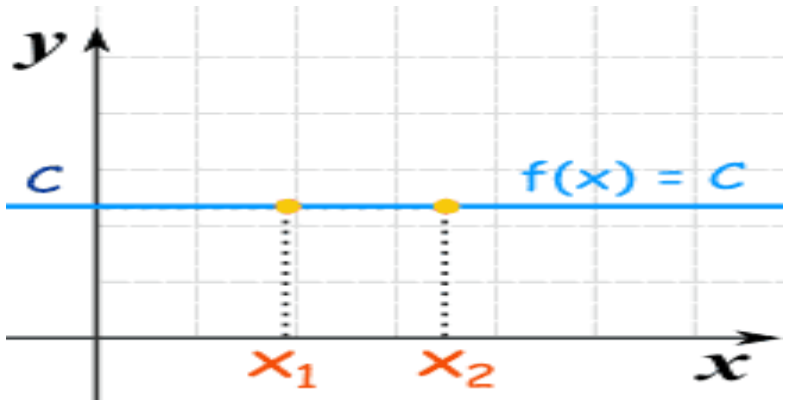
$\frac{x}{a}+\frac{y}{b}=1$ where a and b are the interception on x and y axes respectively.



In the above diagram, the intercepting form of the equation is $\frac{x}{9}-\frac{y}{3}=1$. The x and y interceptions are 9 and 3 respectively.

1.5. Vertical and Horizontal lines or Constant functions:

The constant functions are defined as x = a and y = b, these are the lines parallel to y and x axes respectively.



2.Formation of Linear equations:

Method- 2.1: Method of least squares:

The **least square method** is the process of finding the best-fitting curve or line of best fit for a set of data points by reducing the sum of the squares of the offsets (residual part) of the points from the curve. During the process of finding the relation between two variables, the trend of outcomes are estimated quantitatively. The least-squares method is a crucial statistical method that is practised to find a regression line or a best-fit line for the given pattern. This method is described by an equation with specific parameters. The method of least squares is generously used in evaluation and regression. In regression analysis, this method is said to be a standard approach for the approximation of sets of equations having more equations than the number of unknowns.

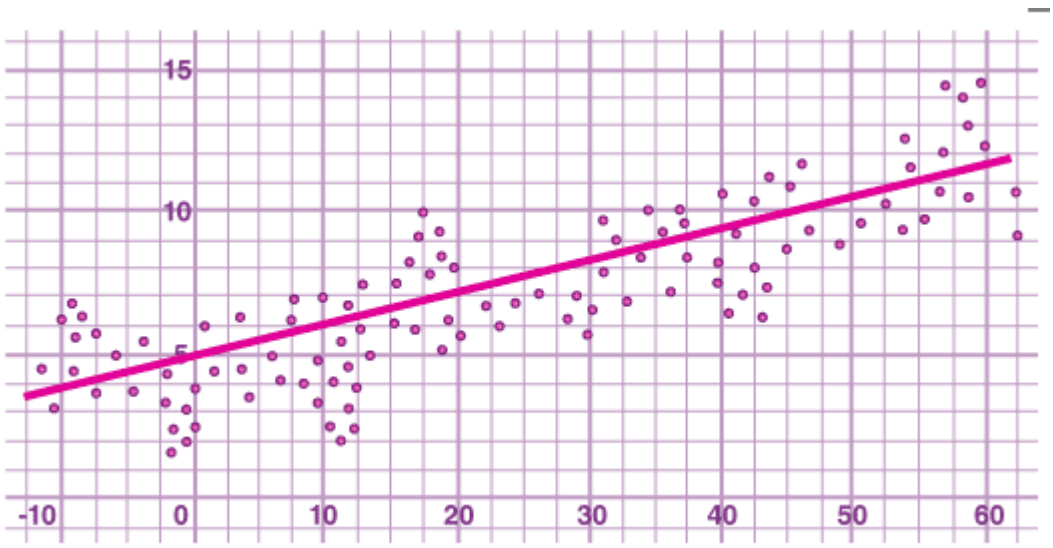
The method of least squares actually defines the solution for the minimization of the sum of squares of deviations or the errors in the result of each equation. Find the formula for sum of squares of errors, which help to find the variation in observed data.The least-squares method is often applied in data fitting. The best fit result is assumed to reduce the sum of squared errors or residuals which are stated to be the differences between the observed or experimental value and corresponding fitted value given in the model.

There are two basic categories of least-squares problems:

- Ordinary or linear least squares
- Nonlinear least squares

These depend upon linearity or nonlinearity of the residuals. The linear problems are often seen in regression analysis in statistics. On the other hand, the non-linear problems are generally used

in the iterative method of refinement in which the model is approximated to the linear one with each iteration. The graph will look like the following:



2.2 Condition for the method of least squares in forming linear equations:

The least-square method states that the curve that best fits a given set of observations, is said to be a curve having a minimum sum of the squared residuals (or deviations or errors) from the given data points. Let us assume that the given points of data are (x_1, y_1), (x_2, y_2), (x_3, y_3), …, (x_n, y_n) in which all x's are independent variables, while all y's are dependent ones. Also, suppose that f(x) is the fitting curve and d represents error or deviation from each given point.

Now, we can write:

$d_1 = y_1 - f(x_1)$

$d_2 = y_2 - f(x_2)$

$d_3 = y_3 - f(x_3)$

…..

$d_n = y_n - f(x_n)$

The least-squares explain that the curve that best fits is represented by the property that the sum of squares of all the deviations from given values must be minimum, i.e:

$$S = \sum_{i=1}^{n} d_i^2$$
$$S = \sum_{i=1}^{n} [y_i - f_{x_i}]^2$$
$$S = d_1^2 + d_2^2 + d_3^2 + \cdots + d_n^2$$

2.3 Formula of Method of Least squares:

Suppose when we have to determine the equation of line of best fit for the given data, then we first use the following formula.

The equation of least square line is given by Y = a + b X

Normal equations are:

$\sum Y = n\,a + b \sum X$

$\sum XY = a \sum X + b \sum X^2$

Solving these two normal equations we can get the required trend line equation.

Thus, we can get the line of best fit with formula y = ax + b, if this is the equation, then the normal equations have to be changed as below:

$\sum Y = a \sum X + n\,b$

$\sum XY = a \sum X^2 + b \sum X$

2.4 Solved Example:

Consider the time series data given below:

x_i	8	3	2	10	11	3	6	5	6	8
y_i	4	12	1	12	9	4	9	6	1	14

Use the least square method to determine the equation of line of best fit for the data. Then plot the line.

Solution:

Mean of x_i values = (8 + 3 + 2 + 10 + 11 + 3 + 6 + 5 + 6 + 8)/10 = 62/10 = 6.2

Mean of y_i values = (4 + 12 + 1 + 12 + 9 + 4 + 9 + 6 + 1 + 14)/10 = 72/10 = 7.2

Straight line equation is y = a + bx.

The normal equations are

$\sum y = an + b\sum x$

$\sum xy = a\sum x + b\sum x^2$

x	y	x^2	xy
8	4	64	32
3	12	9	36
2	1	4	2
10	12	100	120
11	9	121	99
3	4	9	12
6	9	36	54
5	6	25	30
6	1	36	6
8	14	64	112
$\sum x = 62$	$\sum y = 72$	$\sum x^2 = 468$	$\sum xy = 503$

Substituting these values in the normal equations,

$10a + 62b = 72$….(1)

$62a + 468b = 503$….(2)

$(1) \times 62 – (2) \times 10$,

$620a + 3844b – (620a + 4680b) = 4464 – 5030$

$-836b = -566$

$b = 566/836$

b = 283/418

b = 0.677

Substituting b = 0.677 in equation (1),

10a + 62(0.677) = 72

10a + 41.974 = 72

10a = 72 – 41.974

10a = 30.026

a = 30.026/10

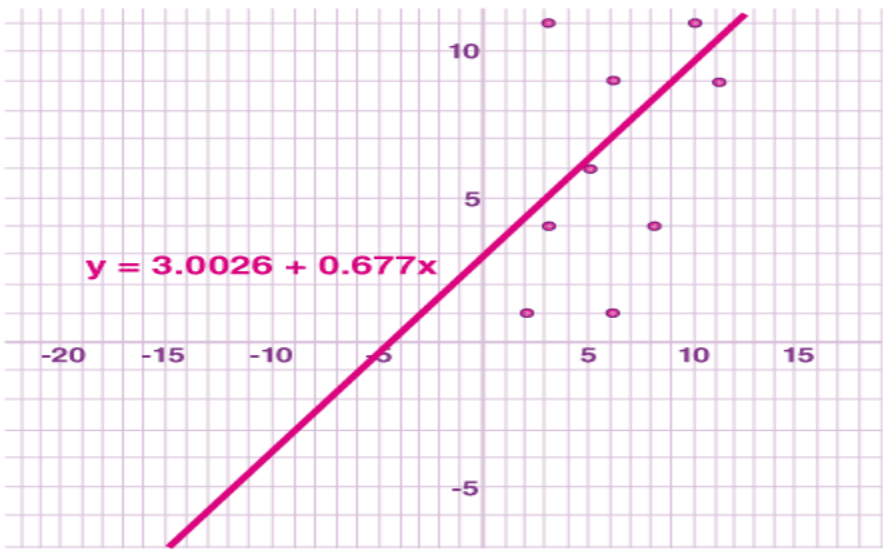
a = 3.0026

Therefore, the equation becomes,

y = a + bx

y = 3.0026 + 0.677x

The trend line is



Now, we can find the sum of squares of deviations from the obtained values as:

d_1 = [4 – (3.0026 + 0.677*8)] = (-4.4186)

d_2 = [12 – (3.0026 + 0.677*3)] = (6.9664)

$d_3 = [1 – (3.0026 + 0.677*2)] = (-3.3566)$

$d_4 = [12 – (3.0026 + 0.677*10)] = (2.2274)$

$d_5 = [9 – (3.0026 + 0.677*11)] =(-1.4496)$

$d_6 = [4 – (3.0026 + 0.677*3)] = (-1.0336)$

$d_7 = [9 – (3.0026 + 0.677*6)] = (1.9354)$

$d_8 = [6 – (3.0026 + 0.677*5)] = (-0.3876)$

$d_9 = [1 – (3.0026 + 0.677*6)] = (-6.0646)$

$d_{10} = [14 – (3.0026 + 0.677*8)] = (5.5814)$

$\sum d^2 = (-4.4186)^2 + (6.9664)^2 + (-3.3566)^2 + (2.2274)^2 + (-1.4496)^2 + (-1.0336)^2 + (1.9354)^2 +$

$(-0.3876)^2 + (-6.0646)^2 + (5.5814)^2 = 159.27990$

3. Practice Problems:

1. Number of man-hours and the corresponding productivity (in units) are furnished below. Fit a linear equation of the form $Y = a + b\,x$ applying the method of least squares. Fit a trend line for the same.

Man-hours	3.6	4.8	7.2	6.9	10.7	6.1	7.9	9.5	5.4
Productivity (in units)	9.3	10.2	11.5	12	18.6	13.2	10.8	22.7	12.7

2. The following table shows the profit of a company in dollars:

x (in years) :	2015	2016	2017	2018	2019
y (in lakhs):	12	19	29	37	45

Fit a trend line using the method of leas squares and also calculate residuals at each of the years. Find the profit of the company in 2020.

3. Apply the method of least squares, fit a straight line of the y = a x + b from the following data:

x : 1	2	3	4	5	6
y : 2.6	5.4	8.7	12.1	16.0	20.2

Plot the line in x y plane and also calculate the value of y when x = 8.

4. Using the method of least squares, fit a straight line of the form y = a x + b from the following data:

x :	0.5	1.0	1.5	2.0	2.5	3.0
y:	05	17	19	14	10	7

5. Using the method of least squares, fit a straight line of the form y = a x + b from the following data:

x:	0	1	2	3	4
y:	1	5	10	22	38

DISCUSSION QUESTIONS:

1. What are the basic forms of straight lines
2. How the geometries of the straight lines be drawn
3. What do you mean by a least square method